Root Cause & Corrective Action Report (RCCA) 10152025

Background 背景

Fabric: 100% Polyester (Fully Oriented Yarn) in Satin structure

Issue: Surface shows “chicken claw” texture / irregular luster marks (表面出現類似雞爪紋的不均勻光澤紋路)

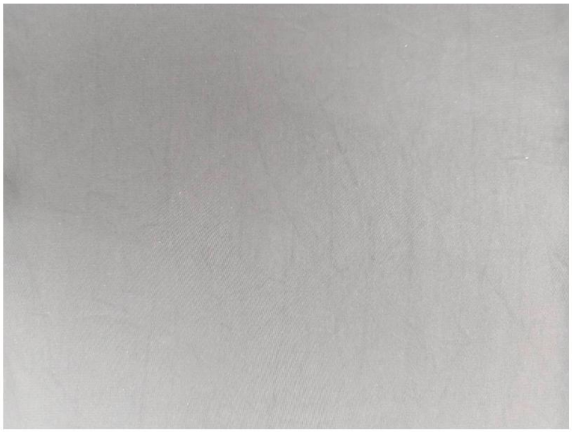
Problem Description 問題描述

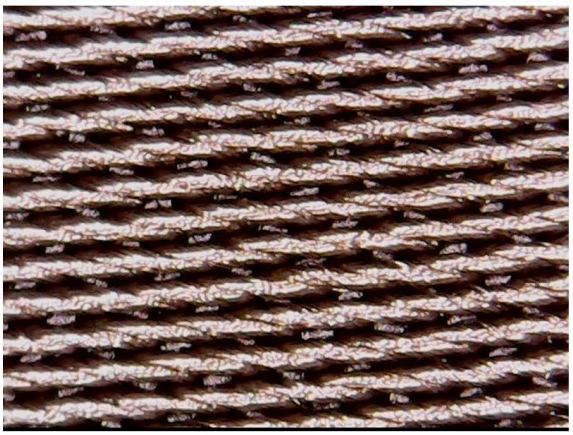
The satin fabric surface shows irregular “chicken claw” - like texture visible under reflection. No significant structural or weave defects are found, but the surface gloss appears uneven, especially when viewed at an angle.

此款聚酯緞面布料在表面反光下可見不規則的「雞爪紋」狀紋理. 組織結構正常，並無明顯織造瑕疵，但光澤表現不均勻，視覺上呈現細微凹凸紋.

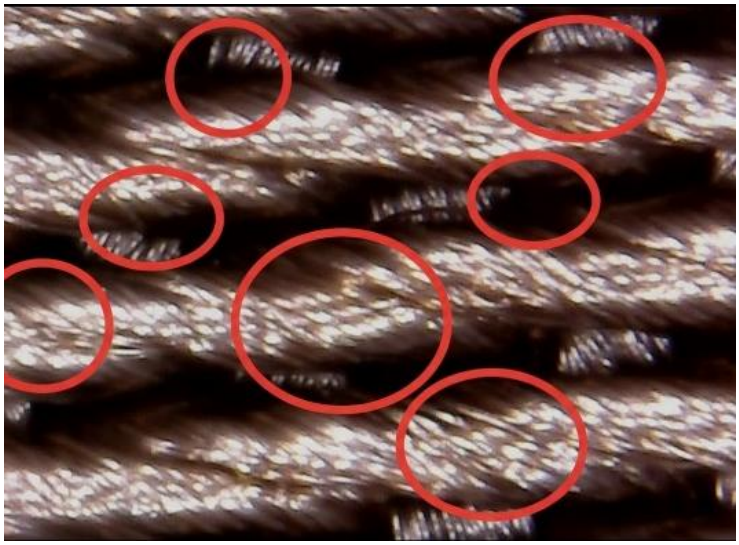
Visual Evidence 視覺證據

Irregular chicken claw and satin structure:





Microscopic yarn close-up:



Possible Root Cause Analysis 根因分析

Microscopic inspection shows consistent weave construction, but filament irregularity along the warp direction. The possible root cause is traced to yarn quality variation rather than weaving or finishing processes.

顯微觀察顯示組織均勻, 但經向長絲表面反光不一, 局部呈現張力不均及微皺現象. 可能的根因判定為原絲品質差異, 非織造或後整處理問題.

Detailed contributing factors 具體影響因素

1. Uneven filament tension / drawing variation (牽伸或張力不穩)

2. Melt spinning or quenching inconsistency (熔融紡絲冷卻氣流不穩)

3. Cone-to-cone lot variation (錠筒批次混用)

Corrective Action 改善措施

Yarn Supplier:

1. Review spinning & drawing process (melt temp, quenching air, draw ratio).
2. Provide U%, CV%, birefringence and luster consistency data.
3. Avoid mixing yarns from different production lots.

Weaving Mill:

1. Use single-lot yarns for warp & weft beams.
2. Maintain stable tension during warping and weaving.
3. Record cone change positions for traceability.

Finishing Mill:

1. Avoid over-tension or uneven heating during heat setting.
2. Test different yarn batches for comparison.

Verification Plan 驗證計劃

Conduct trial weaving using one consistent yarn batch. Compare reflection uniformity and handfeel VS. current batch. Confirm improvement through side-by-side visual and microscopic inspection.

以單一批次原絲重新試織, 觀察表面光澤與手感一致性, 並與現有布匹進行對比檢驗.

Conclusion 結論

This defect originates from filament-level variation during yarn manufacturing (spinning/drawing).

Weaving and finishing amplified the visual effect, BUT are not primary causes.

Corrective focus should be on improving yarn uniformity and batch control.

此不良主要源於長絲生產階段的均勻度問題，織造及後整僅放大了視覺效果。

改善重點為加強原絲均勻性與批次控管。